



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-G2

SYSTEMS INTEGRATION AND

ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS

SPECIFICATION (FRS)

Project Title: Mini App – User Registration & Authentication

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1. Introduction

1.1. Purpose

The purpose of this system is to provide a simple User Registration and Authentication for a web-based mini application.

1.2. Scope

The System will allow:

- New users to Register an account with a Username, Email and Password.
- Registered users to Login using their credentials
- Logged in users to Access the Dashboard, their Profile Page and Protected Pages.
- Users to Logout preventing Access to protected Pages.

System Boundaries:

- The System will only focus on User Authentication, Dashboard Access, Profile Page Access and Protected Page Access.
- The System will not include Advanced features such as Multi-factor Authentication and API based token Authentication.

1.3. Definitions, Acronyms, and Abbreviations

Term	Definition
User	A person who interacts with the System that is either a guest or a registered user.
Dashboard	The page where logged in users will be redirected to upon successful logins. And where users can view personalized information.
Session	This is a Temporary storage that is used to maintain a user's login state.

2. Overall Description

2.1. System Perspective

The system will be a standalone mini web application that provides a simple Authentication workflow. This can be integrated into larger applications in the future.

2.2. User Classes and Characteristics

User Type	Characteristics	Actions
Guest User	Not registered or not logged-in	Register, login
Authenticated User	Successfully logged-in	Access dashboard, profile, protected pages and logout.

2.3. Operating Environment

Hardware: Desktop or Laptop.

Software:

- Web browser
- ReactJS front end and Spring boot backend
- Database: MySQL

Tools:

Code editor (IDE): VS Code

2.4. Assumptions and Dependencies

- The system assumes that the database is operational and accessible.
- Session-based authentication is sufficient for user login and logout.

3. System Features and Functional Requirements

3.1. Feature 1:

Description: Allows new users to Create and Account by providing a Username, Email and Password.

Functional Requirements:

- The system must validate that all input fields are filled.
- The system must check that the email inputted is unique.
- Upon successful registration, the user must be redirected to the Login Page.

3.2. Feature 2:

Description: Allows Registered users to log in with their Email and Password.

Functional Requirements:

- logged in users are redirected to the dashboard
- The system must validate user's credentials against the stored data in the database
- if credentials are correct, then the system creates a session for the user.
- if credentials are invalid, then the system must show an error message.

4. Non-Functional Requirements

Performance: The system must respond fast and not lag. It must respond to Login and Registration requests within a short time.

Security: Passwords must be Hashed and User sessions must be protected.

Usability: The UI/UX must be Clean and Simple.

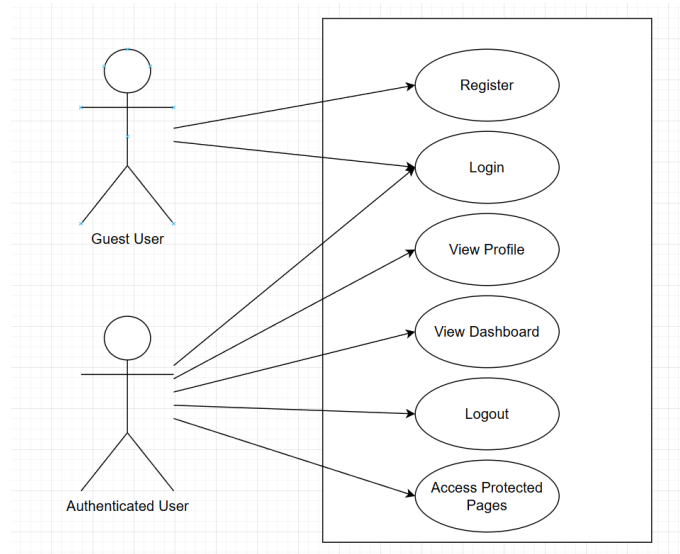
Reliability: The System must be able to handle multiple current users without errors.

5. System Models (Diagrams)

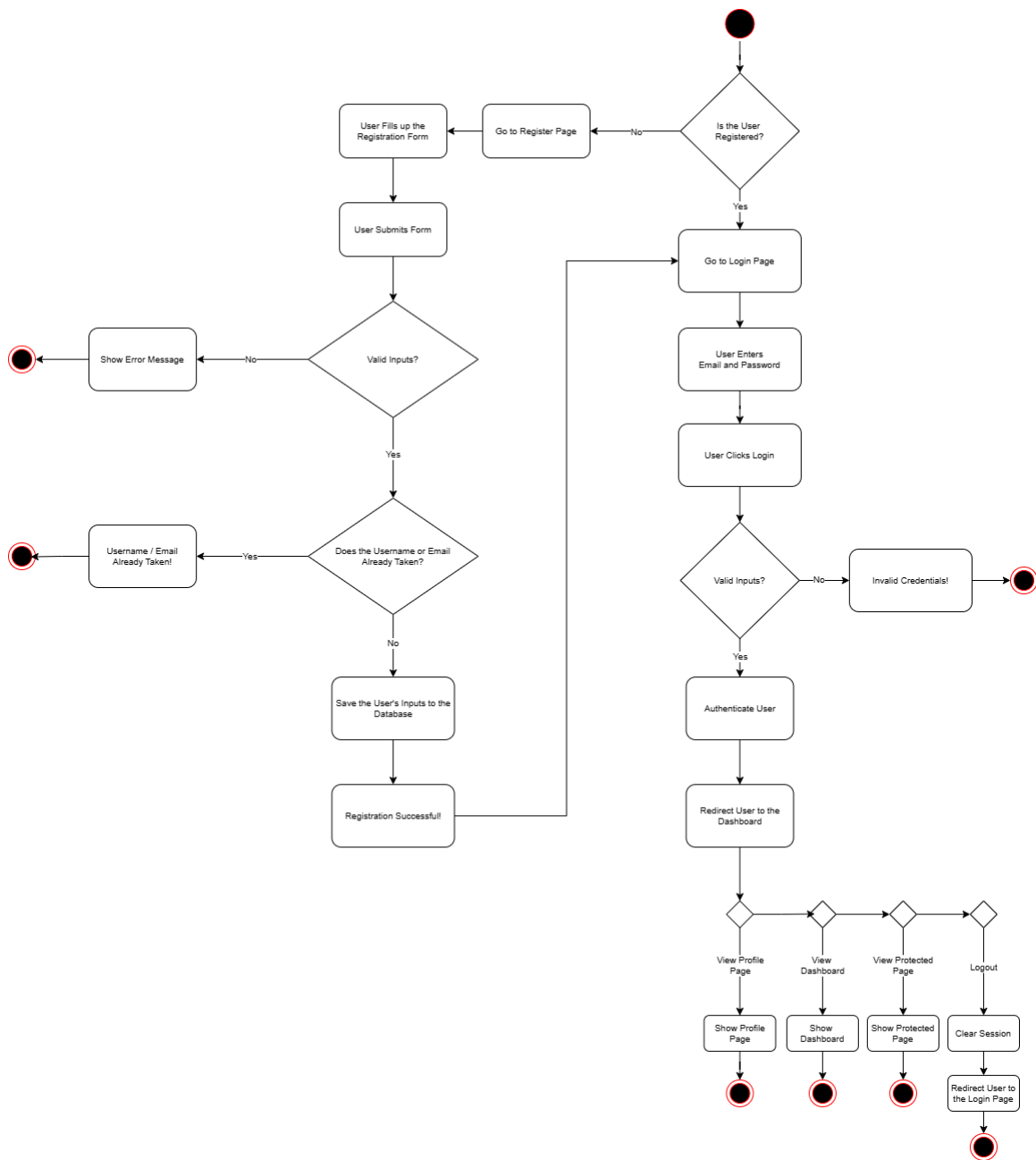
5.1. ERD

User		
PK	<u>userID</u>	<u>int</u>
	username	varchar(255)
	email	varchar(255)
	password	varchar(255)
	dateCreated	date

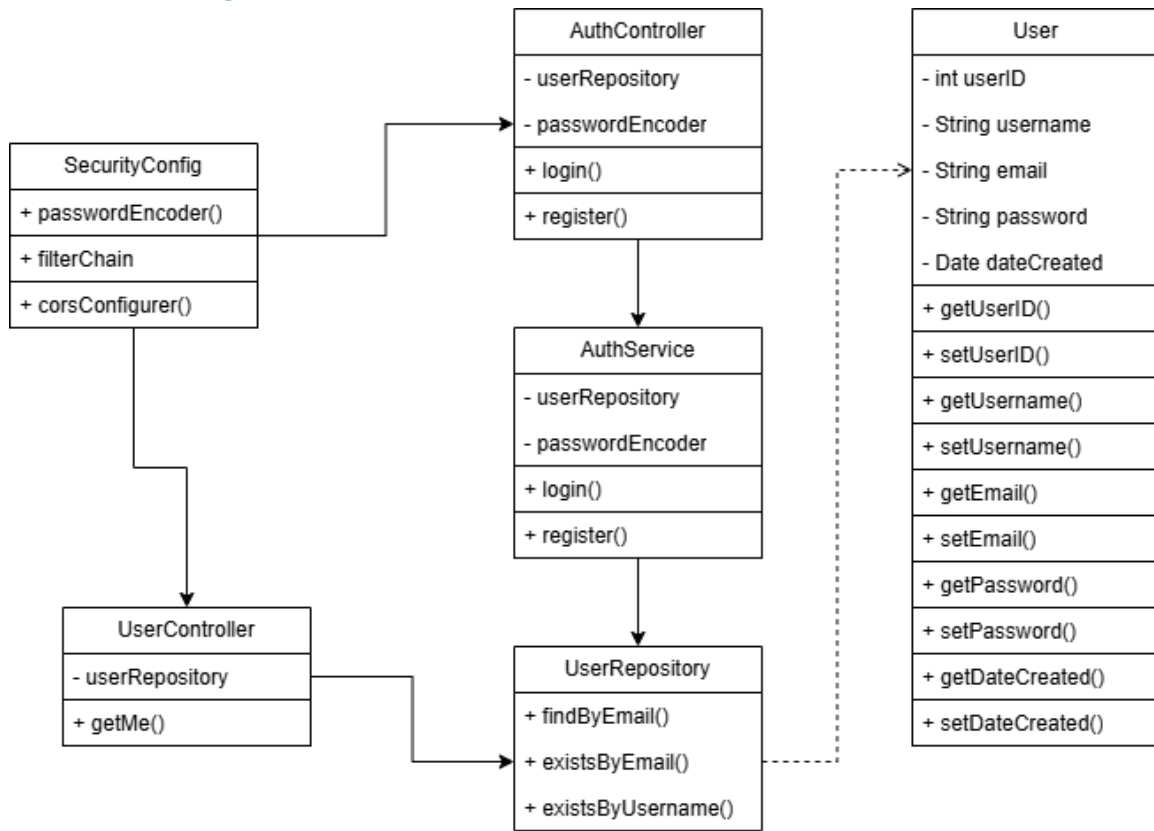
5.2. Use Case Diagram



5.3. Activity Diagram

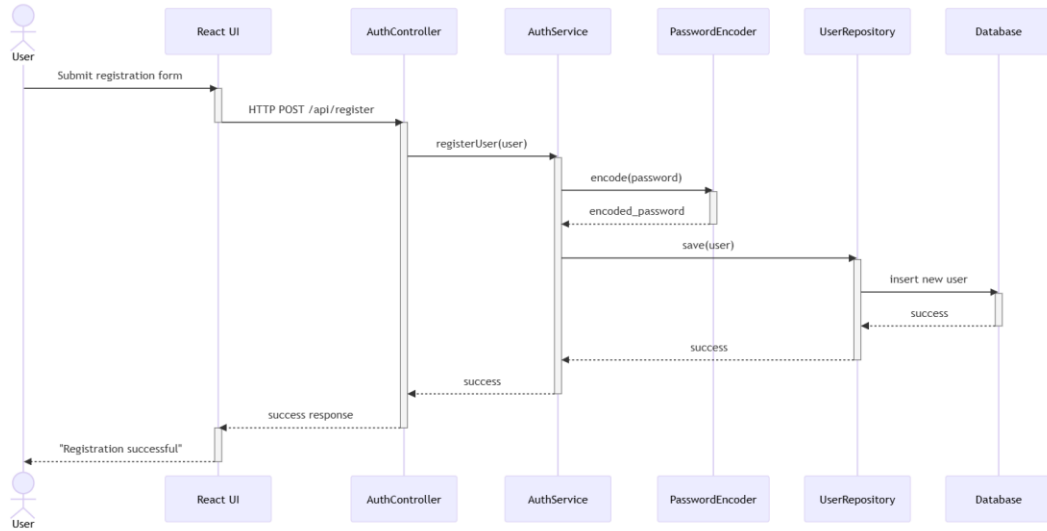


5.4. Class Diagram

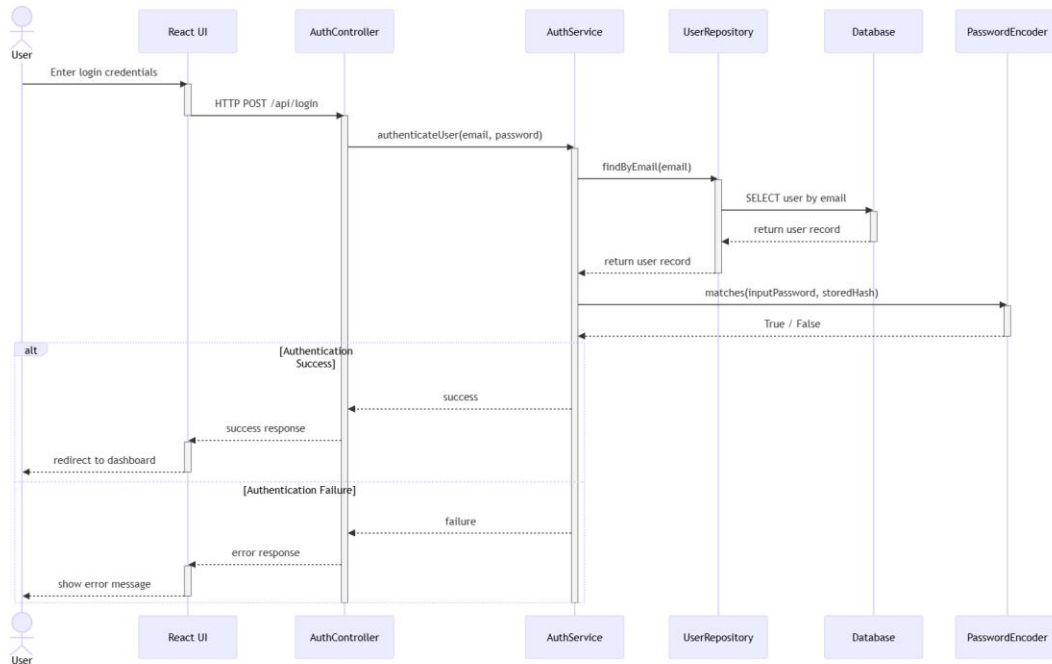


5.5. Sequence Diagram

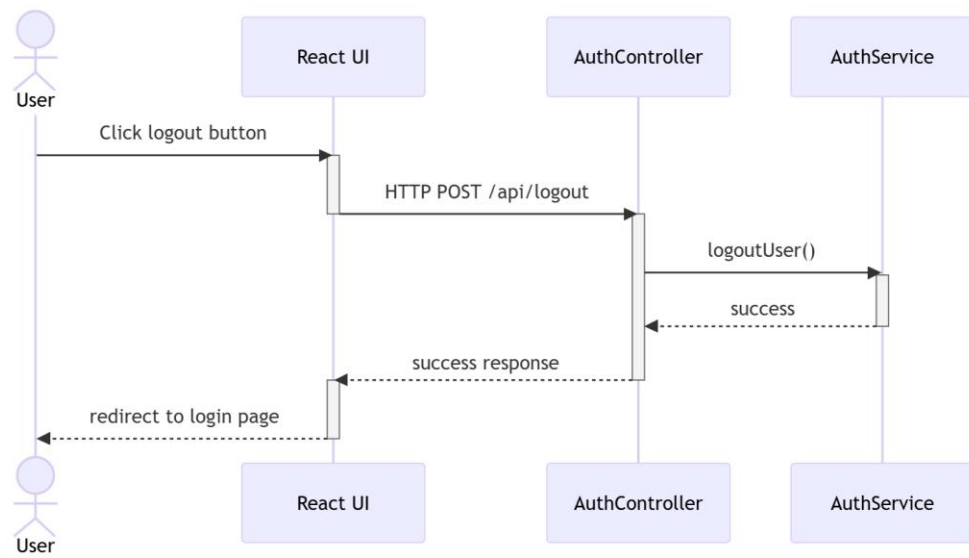
Registration Sequence:



Login Sequence:

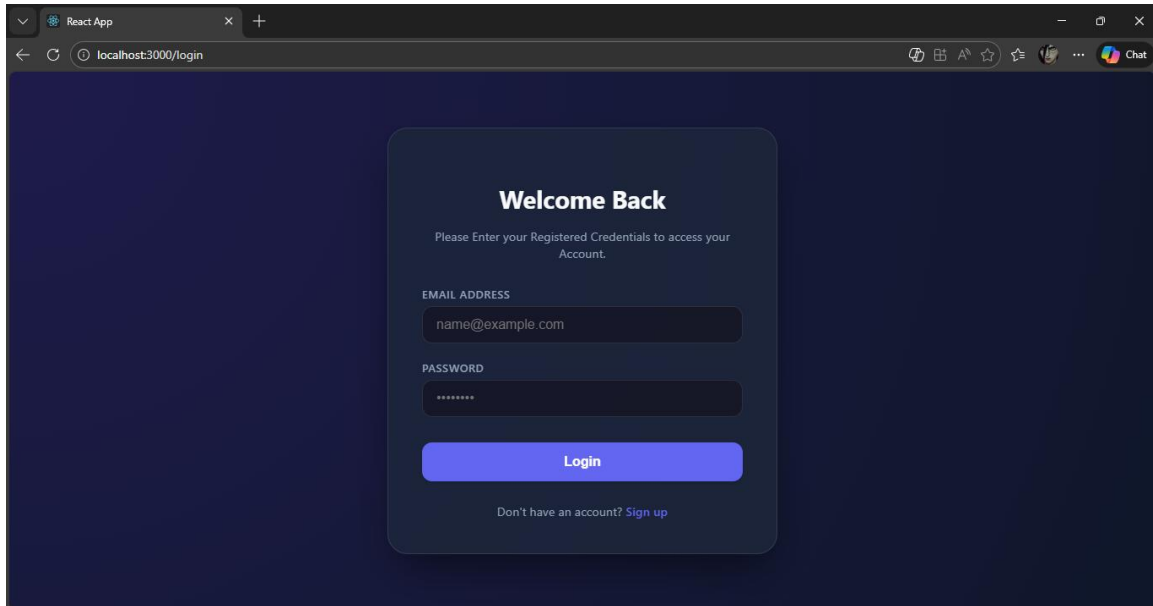


Logout Sequence:



Web UI Screenshots

Login Page:



A screenshot of a web browser displaying a login page. The browser's address bar shows 'localhost:3000/login'. The page has a dark blue background. In the center, there is a light blue rounded rectangle containing the text 'Welcome Back' in bold. Below it, a smaller line of text says 'Please Enter your Registered Credentials to access your Account.' There are two input fields: 'EMAIL ADDRESS' with the value 'name@example.com' and 'PASSWORD' with masked characters. A blue 'Login' button is below the password field. At the bottom, a link says 'Don't have an account? Sign up'.

Welcome Back

Please Enter your Registered Credentials to access your Account.

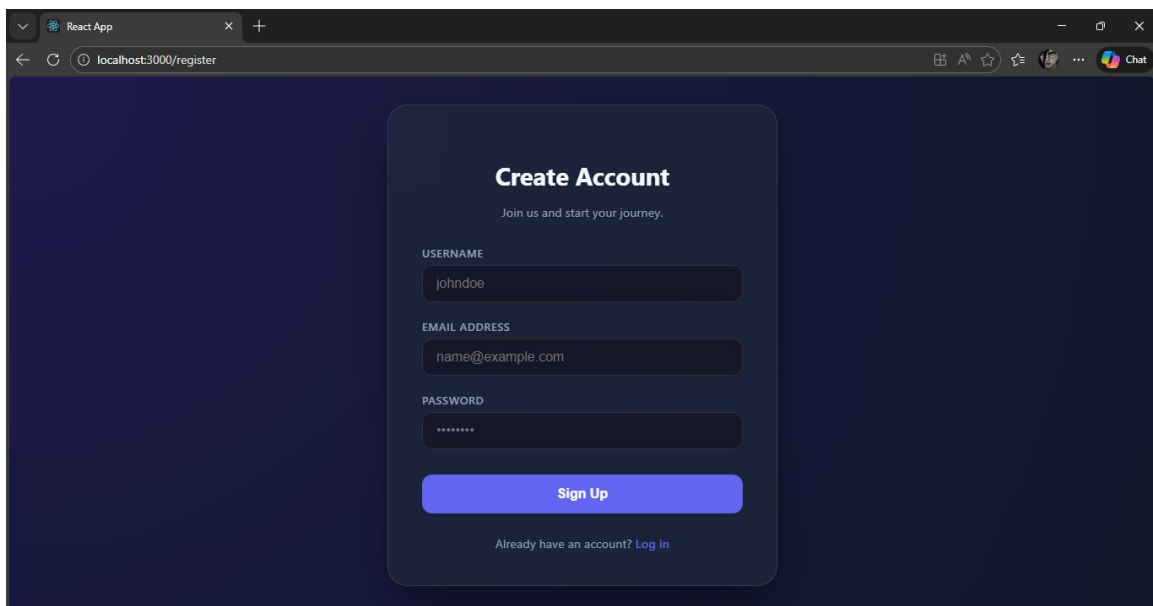
EMAIL ADDRESS
name@example.com

PASSWORD

Login

Don't have an account? [Sign up](#)

Register Page:



A screenshot of a web browser displaying a registration page. The browser's address bar shows 'localhost:3000/register'. The page has a dark blue background. In the center, there is a light blue rounded rectangle containing the text 'Create Account' in bold. Below it, a smaller line of text says 'Join us and start your journey.' There are three input fields: 'USERNAME' with the value 'johndoe', 'EMAIL ADDRESS' with the value 'name@example.com', and 'PASSWORD' with masked characters. A blue 'Sign Up' button is below the password field. At the bottom, a link says 'Already have an account? Log in'.

Create Account

Join us and start your journey.

USERNAME
johndoe

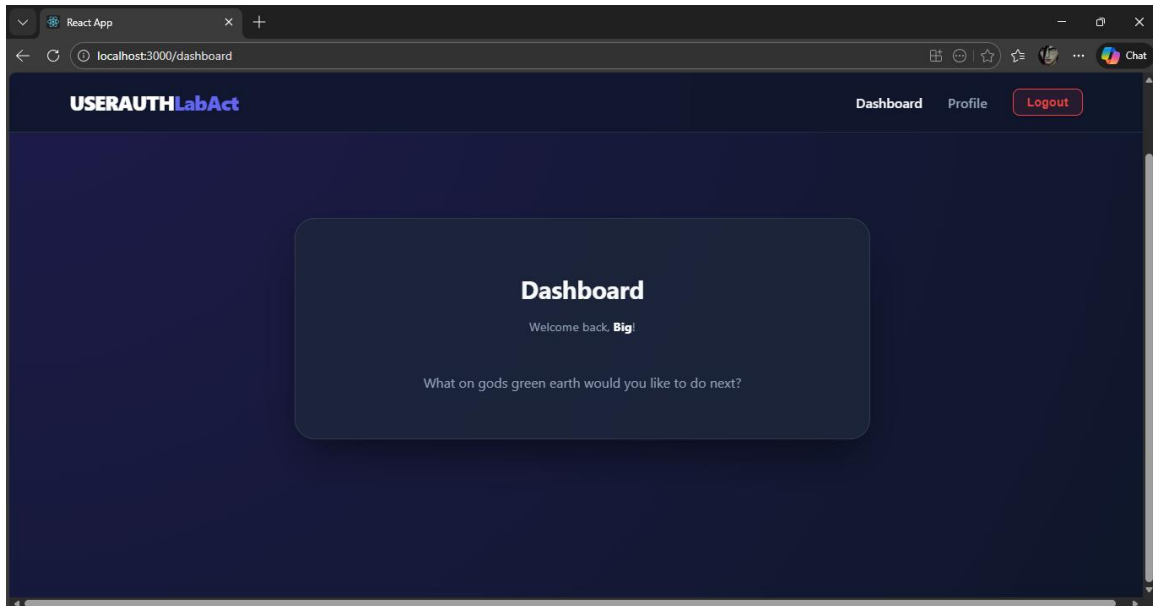
EMAIL ADDRESS
name@example.com

PASSWORD

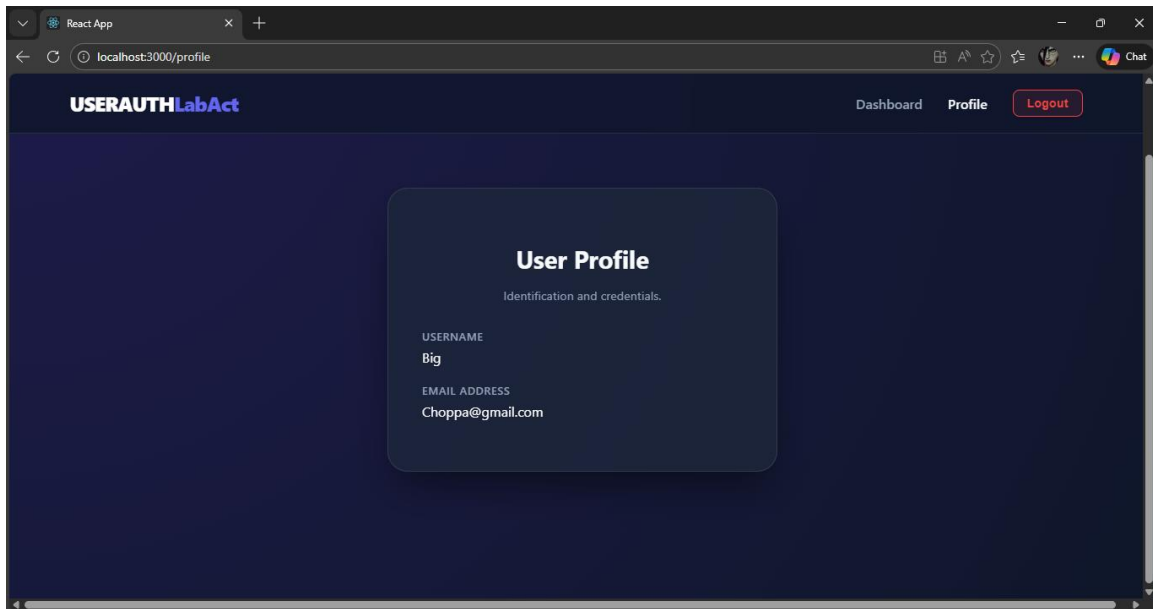
Sign Up

Already have an account? [Log in](#)

Dashboard:

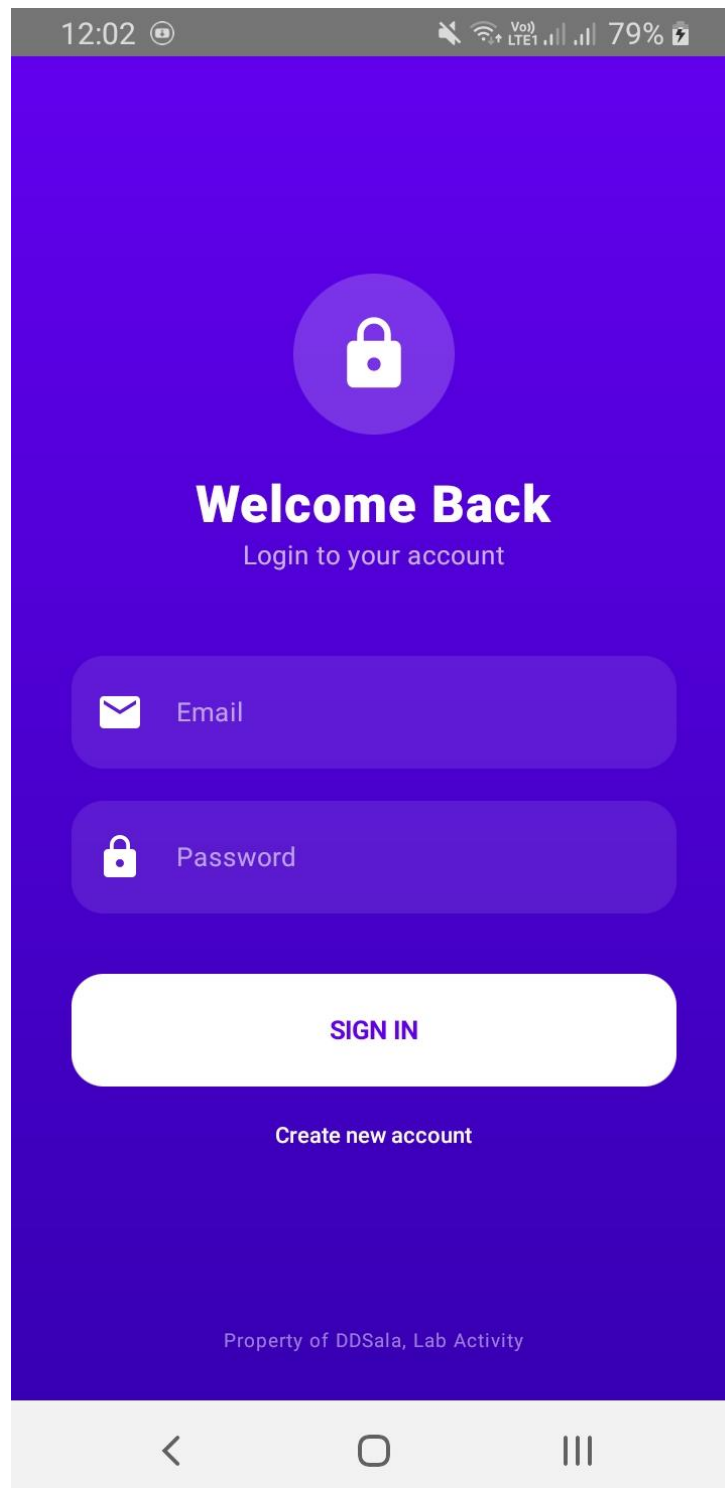


Profile Page:

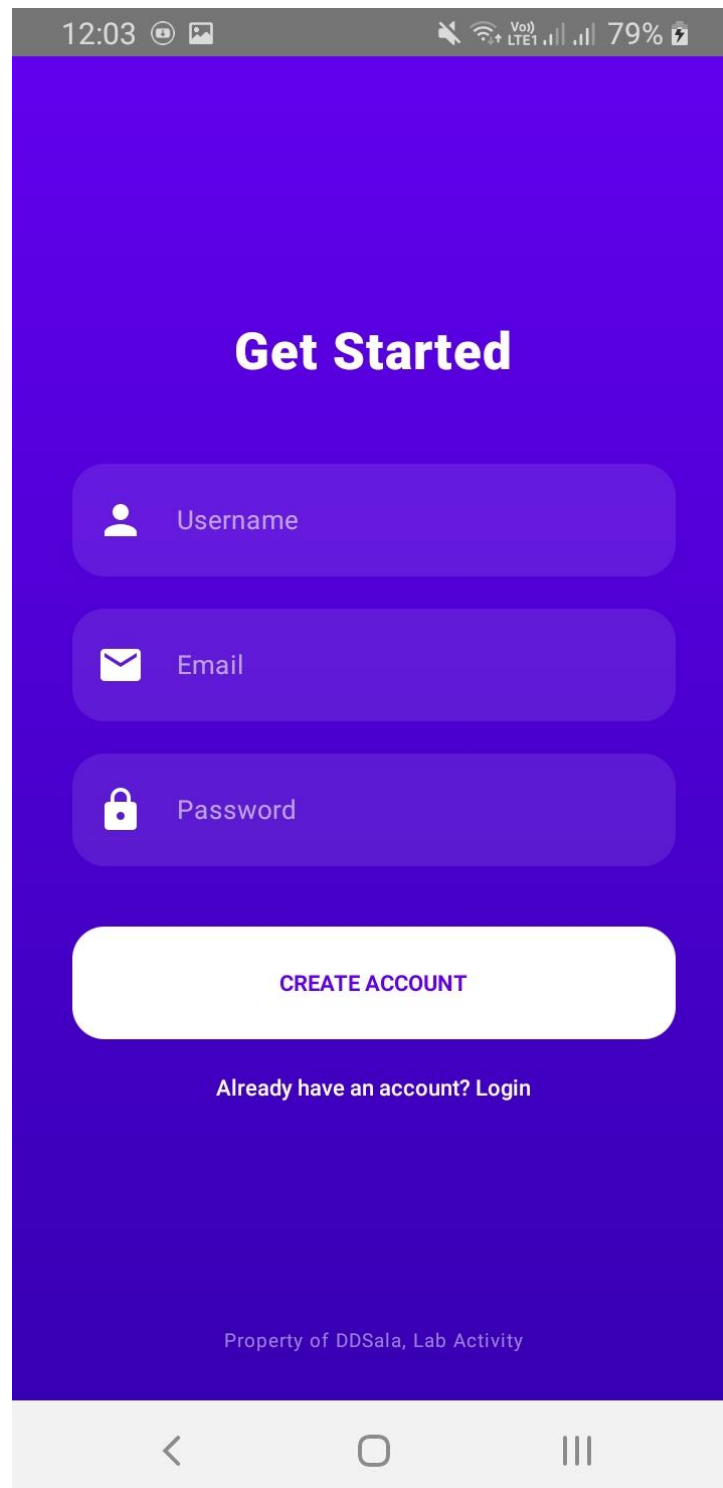


Mobile UI Screenshots

Login Page:



Register Page:



A mobile app register page with a purple gradient background. At the top is a status bar with the time 12:03, signal strength, VoLTE, and 79% battery. The main heading 'Get Started' is centered in white. Below it are three rounded rectangular input fields: 'Username' with a person icon, 'Email' with an envelope icon, and 'Password' with a lock icon. A white 'CREATE ACCOUNT' button is centered below the fields. Below the button is the text 'Already have an account? Login'. At the bottom is a footer with the text 'Property of DDSala, Lab Activity' and an Android navigation bar with back, home, and recents icons.

12:03 79%

Get Started

 Username

 Email

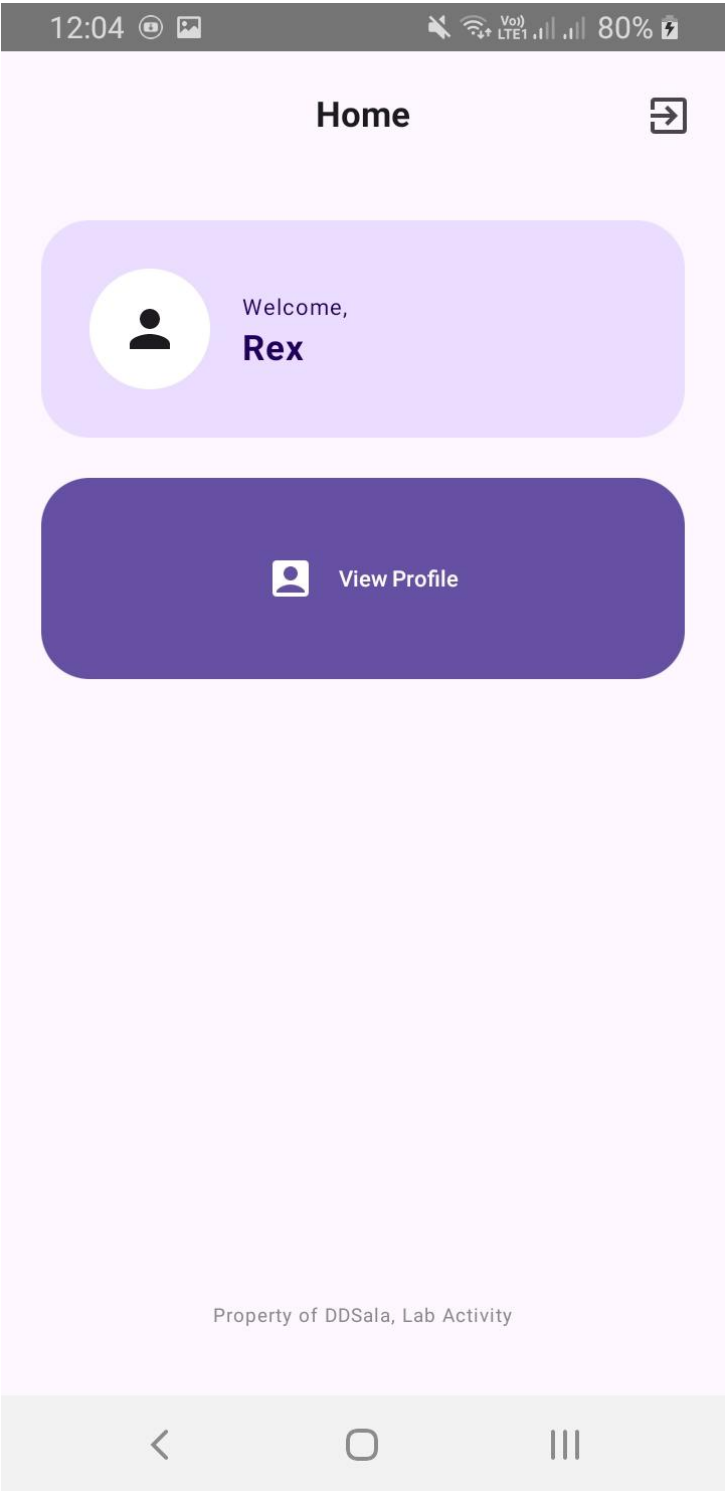
 Password

CREATE ACCOUNT

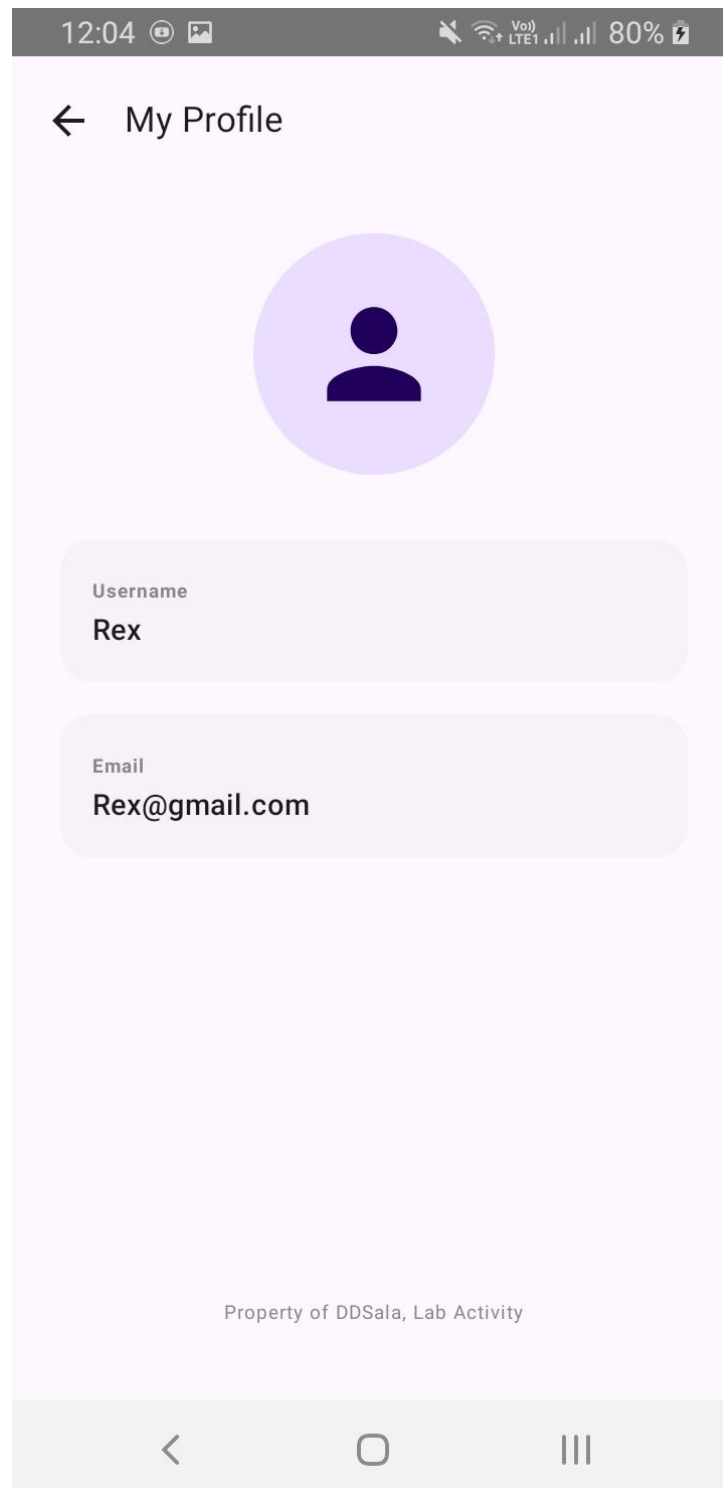
Already have an account? Login

Property of DDSala, Lab Activity

Dashboard:



Profile Page:



6. Appendices

1. Lucidchart. UML Sequence Diagram Guide
<https://www.lucidchart.com/pages/uml-sequence-diagram>
2. TechTarget. Sequence Diagram Definition
<https://www.techtarget.com/searchsoftwarequality/definition/sequence-diagram>
3. Visual Paradigm. Sequence Diagram Tutorial
<https://online.visual-paradigm.com/diagrams/tutorials/sequence-diagram-tutorial/>
4. Research on Functional vs Non-Functional Requirements
arxiv.org/abs/2211.05286